



# Wildfire Q and A Links

- <https://www.nfpa.org/news-blogs-and-articles/blogs/2025/01/15/los-angeles-wildfires-frequently-asked-questions>
- <https://lao.ca.gov/Publications/Report/4952>
- <https://nhmu.utah.edu/sites/default/files/attachments/Wildfire%20FAQs.pdf>
- <https://www.psu.edu/news/research/story/qa-causes-spread-and-solutions-californias-wildfire-crisis>
- <https://www.ucdavis.edu/climate/definitions/wildfire/8-wildfire-questions>
- <https://hsph.harvard.edu/environmental-health/news/health-impacts-of-wildfires-frequently-asked-questions/>
- <https://dec.alaska.gov/air/smoke-faq/>
- <https://www.montecitofire.com/wildfire-prevention-preparedness-frequently-asked-questions>
- <https://azdeq.gov/wildfire-faqs>
- <https://www.worldvision.org/disaster-relief-news-stories/california-fires-facts-faqs-how-to-help>
- <https://sharedsystems.dhsoha.state.or.us/DHSForms/Served/le8626.pdf>
- <https://wildfirepartners.org/general-faq/>
- <https://www.aspenfire.com/wildfire-faq>
- <https://srhd.org/health-topics/environmental-health/wildfire-smoke/air-quality-wildfire-faq>
- <https://www.pscleanair.gov/m/faq?cat=32>
- <https://theodorepayne.org/wildfire-faq/>
- <https://extension.oregonstate.edu/catalog/em-9193-fire-faqs-managing-wildfire-resource-benefit>
- <https://vifr.org/wildland-fire-safety/>
- <https://cityofpetaluma.org/wildfire-faqs/>
- <https://www.wildfirela.org/2025/01/17/before-you-can-go-home-after-the-fires/>
- <https://www.pa.gov/agencies/dcnr/programs-and-services/community-outreach-and-development/wildfire>
- <http://publichealth.lacounty.gov/media/wildfire/faq.htm>
- <https://ndresponse.gov/wildfire-faq>
- <https://coloradosprings.gov/WildfireReady>
- <https://des.mt.gov/Preparedness/Wildfire-Season-Resources>
- <https://www.cawildfirefund.com/sites/wildfire/files/documents/2025/cwf-media-explainer-terms-and-questions-march-2025.pdf>
- <https://www.dir.ca.gov/dosh/wildfire/wildfire-faq.html>
- <https://www.worldvision.org/disaster-relief-news-stories/wildfires-facts-faqs-how-to-help>
- <https://permitsonoma.org/wildfireadapted-faq>
- <https://help.firststreet.org/hc/en-us/articles/5720773111063-Community-Wildfire-Risk-FAQ>
- <https://www.elementalfilm.com/faqs>
- <file:///Users/meenuravi/Downloads/wildfire-smoke-faq-pdf-en.pdf>

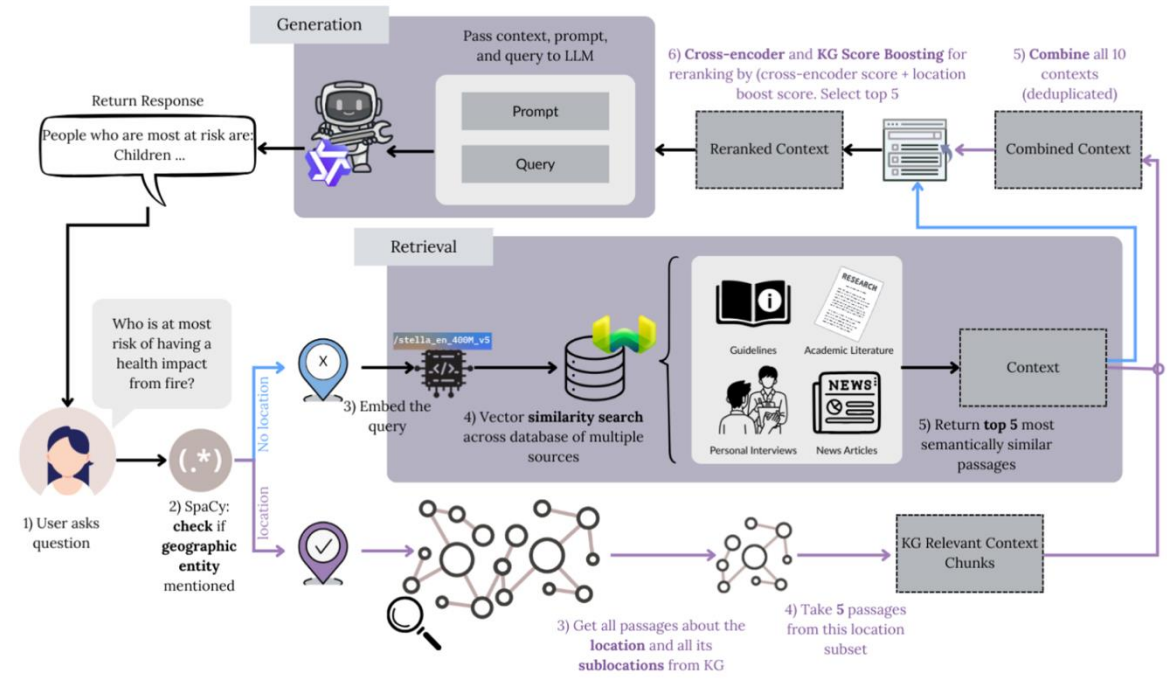
# Research Questions

- How can we measure the impact of geographic granularity in geospatial Q and A?
- What is the most accuracy and effective method of measuring geographic granularity uncertainty
- Is this new UE method more effective than baselines? If so, how much better?

# WildfireGPT

Existing large language models (LLMs) struggle to provide specific and actionable information to the general public

- often generating responses that lack grounding in authoritative sources
- provide generalized advice unsuitable for specific crises
- suggest unreliable resource locators.
- many decision support systems are catered towards professionals with domain knowledge.



**Prompt:** Answer concisely using simple and accessible language as if you are explaining to a novice:

**Query:** Where can I find local resources to help clean my property in LA county after an evacuation?

Ground Truth

FlameAdvisor Response

Local Assistance Centers (LACs) and Disaster Recovery Centers (DRCs) often provide cleanup kits, guidance, and professional referrals for cleaning services. Contact your local public health department for additional resources, including access to protective equipment and cleaning supplies.

You can find local resources to help clean your property in LA county after an evacuation by visiting the [LA County Recovery website](https://recovery.lacounty.gov/returning-after-fire-faq/) at <https://recovery.lacounty.gov/returning-after-fire-faq/>. They provide detailed information about property cleanups and timelines for your neighborhood.

Additionally, there are fire remediation companies that can assist with assessing your home and belongings for any fire or smoke damage, supporting the cleanup process, and advising on safety measures. Remember, residents should avoid cleaning hazardous fire debris themselves to ensure safety.

For further guidance, refer to the LA County Recovery guidelines available at the same website.

If you need immediate assistance, consider calling 2-1-1 for information and referral services. These resources can guide you on how to safely clean up your property post-evacuation.

Aa Text semantically common between ground truth and FlameAdvisor

Aa Text concept present in ground truth but not in FlameAdvisor

Aa Text concept present in FlameAdvisor but not in ground truth

Wildfire related Q and A RAG system for the public (rather than public officials)

- Normal LLM may grab info from reddit or unofficial sources
- RAG with guidelines helps ground answers to authoritative sources
- **Location aware**
  - Sometimes no guidelines or relevant passages about a specific location so how can we still offer response but make it clear/known to user- give a trustworthy index or score

# Geographical uncertainty quantification

- Geographic granularity is still underexplored dimension of RAG uncertainty measures
  - Simple distance-based scoring may be insufficient
    - You can't just retrieve sources and assume high confidence
  - Add environmental context similarity
  - If no documents exist at the city/county level, the RAG should retrieve documents from broader levels and lower its confidence proportionally and vice versa.
- Query-dependent geographical similarity scoring for RAG uncertainty
  - Understand how RAG systems assess retrieved passage relevance
- The dataset - wildfire Q&A benchmark with spatial metadata attached

# Tasks

- Compile a more expansive dataset of geographic related wildfire QA
- Identify best method of extracting locations from passages
- Label questions based on type of question (i.e., policy, health, informational etc.)
  - Some questions like evacuation packing list is same is all locations
- Identify baseline UE measures
- Identify and come up ideas for best UE measures
- Identify ways for accounting for sibling relations- different county within same
- Maybe doing administrative hierarchy for policy questions and geographic uncertainty for other questions

- Sigspatial
- Cikm
- Ickg (ieee international conference on knowledge graph)

# April 10 2026

- Address reliability of news sources
  - How?
    - I filter news articles to commonly reputable domains (.gov, .org, and major)
- Instead of chunking based on set number of tokens, I use semantic splitter so that similarly texts and sentences stay together
- GeoExtraction
  - Looked at 2 geo extractors: FlairNER and bertbase NER
    - FlairNER seemed to perform much better
  - **Issue to discuss:**
    - For example I have a text like “Unlike New Mexico, Texas wildfires are much more severe” How can I get Texas instead of both New Mexico and Texas
    - NER methods get both. I think the other option is LLM
    - 500 manually labeled data ner and ai model compare on manually labeled data as a way to validate

| Category                  | Domains                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Sources           | apnews.com, reuters.com, afp.com, upi.com                                                                                                                                                                                                                           |
| National U.S. Newspapers  | nytimes.com, washingtonpost.com, wsj.com, usatoday.com, npr.org                                                                                                                                                                                                     |
| Major Regional Newspapers | chicagotribune.com, latimes.com, bostonglobe.com, dallasnews.com, sfchronicle.com, denverpost.com, seattletimes.com, startribune.com, tampabay.com, miamiherald.com, houstonchronicle.com, inquirer.com, baltimoresun.com, sandiegouniontribune.com, oregonlive.com |
| Major International       | theguardian.com, ft.com, economist.com, abc.net.au, cbc.ca                                                                                                                                                                                                          |
| Investigative / Long-form | propublica.org, theatlantic.com                                                                                                                                                                                                                                     |
| Fire / Disaster Specific  | wildfiretoday.com                                                                                                                                                                                                                                                   |
| Government                | .gov, .org                                                                                                                                                                                                                                                          |



# April 10 2026

- I went through the questions answer dataset and manually labeled 2 things
  - 1. Category
    - "1. General Wildfire behavior: causes, spread, fire science, smoke behavior, seasons, risk factors.\n"
    - "2. Home and property: protecting homes, defensible space, structures, insurance for homes, property damage.\n"
    - "3. Codes and Regulations: laws, building codes, permit rules, compliance requirements, official regulations.\n"
    - "4. Specific fire event: questions about a named wildfire, a particular incident, location-specific active or past fire.\n"
    - "5. Community: questions about how the community can help, community-level impacts.\n"
    - "6. Land: forests, vegetation, soil, erosion, watersheds, land management, acreage, habitat.\n"
    - "7. Finance: grants, loans, compensation, taxes, financial aid, business loss, economic cost.\n"
    - "8. Wildfire response and prevention: evacuation, preparedness, mitigation, fuel reduction, emergency response.\n"
    - "9. Wildfire recovery: debris removal, rebuilding, restoration, post-fire recovery steps, returning after fire.\n"
    - "10. Health and Safety: smoke exposure, breathing, injury, health risks, safety precautions for people.\n"
    - "11. Informational: general factual or definitional questions that do not fit the other categories strongly.\n\n"
  - 2. Does it require knowledge of exact location
- Then, I repeated the same process but asked llm to label it

```
"role": "system",
"content": (
    "You are given a question-answer pair. "
    "Determine whether answering the question correctly
    requires "
    "knowledge of a specific geographic location, such as a
    city, county, state, region, or named fire area.\n\n"
    "Answer YES if the question depends on place-specific
    facts.\n"
    "Answer NO if the question can be answered generally
    without location-specific knowledge.\n\n"
    "Rules:\n"
    "- Return only YES or NO.\n"
    "- Do not explain your choice."
```

```
"role": "system",
"content": (
    "You are given a question. "
    "Classify the QUESTION into exactly one category from the
    list below.\n\n"

    "Rules:\n"
    "- Use the answer only as supporting context.\n"
    "- Classify based mainly on the question.\n"
    "- Choose exactly one category.\n"
    "- Return only the category name.\n"
    "- Do not explain your choice."
```

# April 10 2026

- To do next:
  - For questions that are location dependent, I want to get answers for other locations
  - My goal is to show that while the same general information may be similar, specific details like phone numbers, who to contact – those are the parts that are different and this is where semantic similarity is not enough
    - Is llm as judge the way to go?- mention the types of question and evaluate like this- give instructions
    - Based on the change of the objective- if goal is show general formation
- Graph based distance
  - Location query location each retrieved location and finding

# Jurisdiction-sensitive retrieval in RAG Intro

- jurisdiction sensitivity is a latent property of questions. Geographic mismatch matters disproportionately for location-sensitive questions.
  - Someone dealing with a wildfire in Grand County needs *some* answer, not none. But they also need to know "this is city-level policy, your county may differ."
    - return the best available answer with a calibrated warning about how far the jurisdiction is from what was asked.
- A query is jurisdiction-sensitive if the correctness or applicability of an answer depends on the governing geographic or administrative scope.
  - semantic relevance and geographic proximity do not capture applicability. If open burning rules are lenient in Colorado, they may be illegal in Utah despite them being proximally close
- **UE: the distance between what jurisdiction the question needs and what jurisdiction the answer comes from.** Small distance is answer with low warning. Large distance is answer with high warning.
  - low risk: exact jurisdiction match
  - moderate risk: nested or governing jurisdiction
  - high risk: sibling or unrelated jurisdiction
  - very high risk: no grounded geographic evidence

# Dataset Info

when risk is high, do warning, fallback, or abstention  
policies reduce harmful answers compared with a normal  
RAG baseline

- General education / health / preparedness -select answer
- Local policy / permit / restriction- select answer\_with\_warning
- Live incident status / evacuation / current closures- select abstain

|        |          |        |      |       |                   |                       |                                     |               |                    |
|--------|----------|--------|------|-------|-------------------|-----------------------|-------------------------------------|---------------|--------------------|
| Source | Question | Answer | City | State | jurisdiction_type | intended_jurisdiction | expected_behavior_if_no_exact_match | Question_Type | Location Dependent |
|--------|----------|--------|------|-------|-------------------|-----------------------|-------------------------------------|---------------|--------------------|

# How to prove the risk metric adds value beyond exact-match filtering- it ranks mismatch and supports better decisions under imperfect retrieval.

- Model Baselines: bge small en+gemma, bge small en+gpt-oss, bge base en+gemma, bge base en+gpt oss
- Method Baselines: Similarity-only normal RAG, Exact metadata filter, jurisdiction rerank, my method
- Evaluation levels: retrieval-level, answer-level
- Ablations: graph distance only, granularity only (only looking levels), full metric
- Judging: semantic similarity, LLM-as-judge (good at reasoning mistral, gpt oss reasoning focused) for grounded, retrieval quality etc.
  - How close retrieved passages are to question
  - How correct answer is to the ground truth (cross encoder to label to label support or opposes gt (entailment task we can check using cross encoder))
- Error buckets: exact miss, sibling near-miss,

| Method           | Purpose                                        |
|------------------|------------------------------------------------|
| Similarity-only  | Just semantic similarity                       |
| Exact filter     | Uses location keyword match only               |
| Graph-only       | Graph distance                                 |
| Granularity-only | Keeps only exact jurisdiction level docs       |
| Full method      | Uses graph distance + granularity + similarity |

## Plan:

1. retrieve top-k with each method
2. for every retrieved passage, get sem sim score, jurisdiction, and the graph distance
3. compare methods on whether they rank better evidence higher
4. then generate answers
5. Evaluate answer / warning / abstain behavior, with LLM judge and some human checks.

## show:

- filtering alone fails when exact docs are missing
- similarity alone retrieves wrong-but-similar jurisdictions
- this method handles mismatch better by deciding when to warn or abstain, or by preferring lower-risk evidence

# Notes and thoughts

- Phrase it as risk-aware retrieval for scope-sensitive questions
  - # of questions? Loc specific- around 250 non location 250 (try to keep both around same size)
  - How to choose topk- threshold
    -
  - I think we can phrase it as:
    - We investigate retrieval for geographically scope-sensitive questions using wildfire information systems as our evaluation domain, where semantically similar documents may be unsafe or inapplicable because they come from the wrong jurisdiction or authority level. We propose an applicability-risk metric that improves evidence-based ranking and supports selective answering, warning, and abstention under imperfect retrieval.
  - missing or inconsistent parent-child relations – used wikidata to build initial mapping and manually prune missing child
    - differences across countries in administrative structure,
    - ambiguous place names
    - places that belong to multiple relevant structures,
    - governing bodies that do not align cleanly with geography- ask the user/clarification
- Solutions
  - Wikidata for hierarchy and aliases
  - Who's On First or GeoNames for place identity and hierarchies
  - OSM or GADM for boundaries and overlap
  - official registries for functional authorities like school districts, tribal lands, utilities, fire districts
  - Add to limitations

- Butte County has a unique geography. It extends from the floor of a valley into a high-mountain region. The landscape is made up of ridge lines that separate the county's communities. Canyons split the county into seven regions, the jagged terrain limiting access across the ridges. Communication and outreach are difficult and time-consuming efforts for residents and officials. The physical differences in the county caused the need for custom strategies for each community.
- <https://www.fema.gov/case-study/ashes-paradise-community-plans-forged-wildfire>

Week of 04/19



# Resolving Some Limitations

- Resolve a place name into its administrative hierarchy (city → county → state → region → country) using Wikidata
  - These 4 are the main levels referenced in these types of questions
  - Not much in municipality level, especially in US
- 4 prioritized disambiguation methods for ambiguous place names like "Springfield."
  - Explicit state: if state is provided, appends it to the query for direct lookup
  - Context: extracts state mentions from surrounding text via flair NER
  - Population: picks the most populous US candidate from Wikidata
  - Sitelinks: fallback to the most Wikipedia-referenced candidate
    - Level of confidence based on number of queries asked about that location
    - Based on this ask
- For natural area, wikidata uses:
  - Mouth of river, summit of mountain
  - In practical use we would ask user for follow up
- Alias names from wikidata
  - E.g. the sunshine state for florida
- For tribal lands, wikidata uses
  - Maps to all the states that the land covers
- Add questions without clarifying location and test how well the clarifying questions are.

# Get all locations from chunked passages

- Use flair ner
- Gets a list of mentioned locations
- Uses state folder for breaking ambiguities
- Get aliases also
- From list use llm to get what the passage is about

# Geo Extraction Steps

- Load pkl files of semantic nodes (news + pdf)
- Run `process_pkl_batched` which batches nodes, creates Flair sentences in parallel, predicts NER tags, extracts LOC entities with confidence  $\geq 0.85$ , normalizes location variants, and saves checkpoints in background
- If no locations found for a node, fall back to folder name
- Save

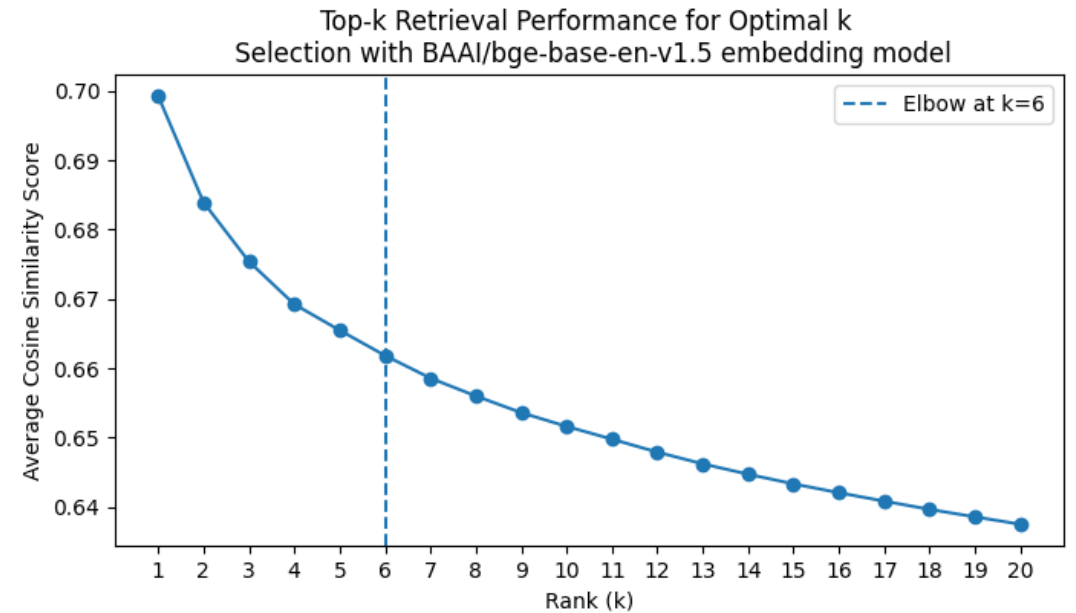
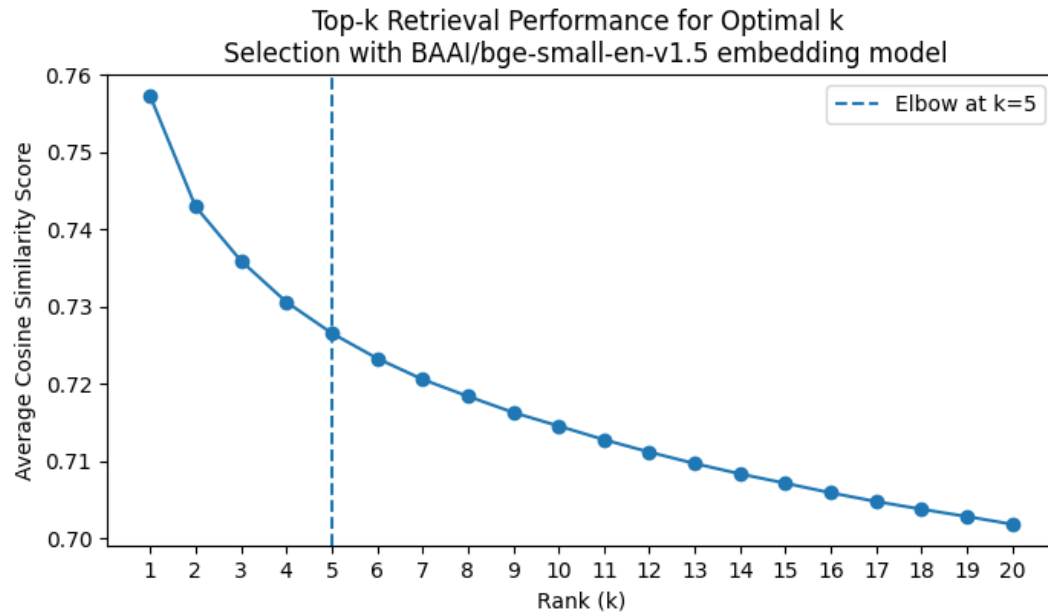
Week of 04/26

# Selecting Appropriate top-k Value

Plot the quality score of answer vs k value

How relevant answer is to k value

Metric that evaluates the quality of answer (truthfulness, readability, Groundedness, uncertainty)



Final selection k=6

# Result with bge small+gpt oss

|                        | location_group | Location-dependent | Non-location |
|------------------------|----------------|--------------------|--------------|
| metric                 | method         |                    |              |
| same_scope_rate        | exact_match    | 0.835              | 0.948        |
|                        | vanilla        | 0.518              | 0.660        |
|                        | granularity    | 0.559              | 0.794        |
|                        | graph_match    | 0.873              | 1.000        |
|                        | risk_aware     | 0.615              | 0.829        |
| mean_answer_similarity | exact_match    | 0.557              | 0.633        |
|                        | vanilla        | 0.586              | 0.672        |
|                        | granularity    | 0.566              | 0.650        |
|                        | graph_match    | 0.551              | 0.589        |
|                        | risk_aware     | 0.563              | 0.642        |
| abstention_rate        | exact_match    | 0.680              | 0.448        |
|                        | vanilla        | 0.605              | 0.359        |
|                        | granularity    | 0.659              | 0.417        |
|                        | graph_match    | 0.687              | 0.574        |
|                        | risk_aware     | 0.663              | 0.438        |

| Method               | Same-scope        | Answer similarity | Abstention | Insight                                                        |
|----------------------|-------------------|-------------------|------------|----------------------------------------------------------------|
| Exact keyword match  | 0.835             | 0.557             | 0.680      | Precise when exact documents exists, but often fails/abstains  |
| Similarity-only      | 0.518             | 0.586             | 0.605      | Strong answer quality, but weakest geographic reliability      |
| Granularity-only     | 0.559             | 0.566             | 0.659      | Matching level alone helps only slightly                       |
| Graph match          | 0.873             | 0.551             | 0.687      | Strongest geographic safety, but sacrifices answer quality     |
| Full geo-first       | 0.615             | 0.563             | 0.663      | Best balance between geographic reliability and usable answers |
| Normal Cross Encoder | Currently running |                   |            |                                                                |

# LLM as a judge metrics

For each question, I ask to pick one and give reason

Using nemotron—3-super

| Metric |                               | Definition                                                                                                                                            |
|--------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <b>Accurate Decision Rate</b> | <b>Measures Reliability:</b> The ratio of queries where the system either provided a correct answer or accurately signaled its own limitations.       |
| 2      | <b>Coverage/Attempt rate</b>  | The percentage of queries where the system attempted a factual answer. High coverage means lower abstention rate.                                     |
| 3      | <b>Supported Answer</b>       | <b>Measures Accuracy:</b> The answer is factually correct and applies precisely to the requested jurisdiction.                                        |
| 4      | <b>Wrong Confident</b>        | <b>Measures Critical Failure:</b> The system provided a confident answer that was factually wrong or applied to a mismatched jurisdiction.            |
| 5      | <b>Appropriate Warning</b>    | <b>Measure graceful fallback:</b> The system correctly identified that evidence was partial (e.g., state-level only) and warned the user.             |
| 6      | <b>Appropriate Abstention</b> | <b>Measures Safety:</b> The system correctly refused to answer because evidence was missing or geographically irrelevant.                             |
| 7      | <b>Over-Abstention</b>        | <b>Measures Efficiency Loss:</b> The system refused to answer even though relevant jurisdictional evidence was present in the context.                |
| 8      | <b>Avg. Evidence Coverage</b> | <b>Measures Retrieval Quality:</b> The average number of passages (0–6) per query that were both factually and geographically relevant.               |
| 9      | <b>95% CI</b>                 | <b>Measures Statistical Significance:</b> The range within which the true metric mean lies, calculated via bootstrap resampling with 1000 iterations. |

# Next Steps

- 5 csv files with llm generated answers and ground truths
- Want to look at How correct answer is to the ground truth
  - cross encoder to label to label support or opposes gt
  - (entailment task we can check using cross encoder)
- If we ask questions without clarifying location – how can we ask clarifying questions and test how well the clarifying questions are.
  - Question: What wildfire mitigation rules apply in Springfield?
  - System should not immediately retrieve from the most prominent Springfield. It should detect that Springfield has multiple plausible jurisdictions and ask:
    - Which Springfield do you mean: Springfield, Illinois; Springfield, Massachusetts; Springfield, Oregon; or another Springfield?
  - Ambiguous place names introduce a different form of geographic risk in that the system may retrieve highly relevant evidence for the wrong jurisdiction prior to applying meaningful graph-distances. In such scenarios, the appropriate behavior is not to rank or abstain immediately, but to ask a clarification question. We demonstrate this with ambiguous toponyms such as “Springfield,” where multiple U.S. jurisdictions are plausible. This case study shows how geo-risk can be used not only for retrieval-time filtering, but also for deciding when the user’s geographic intent is underspecified.



# Case Studies

- Case 1: Exact keyword filtering fails
- Case 2: Similarity-only retrieves wrong jurisdiction
- Case 3: Full geo-first warns or abstains
- Case 4: Ambiguous toponym clarification
- Experiment section- evaluating models on quality, uncertainty at answering location specific qu, how it abstains
  - Evaluation criteria/settings
- For each- eval matrix (3 tables)
- How evaluation matrix is created using llm as judge

FLLM Idea

# Idea behind localizing geospatial decision making in LLMs

- Current methods to control LLM behavior rely on prompt engineering.
- However, prompting is surface-level, token-expensive, and easily manipulable by the model's pre-trained biases.
- If we understand *where* geography/geospatial information is stored, we can make these models more interpretable and reliable for high-stakes natural hazard mitigation tasks
- Geospatial information can be ambiguous with linguistic data. We don't know if a model understands "Paris" as a location or just a high-probability token following "Eiffel Tower."
- We lack a precise map of which layers are responsible for transforming a place name into a spatial decision
- Models sometimes hallucinate proximity (e.g., say two distant cities are neighbors) because their internal spatial vectors are poorly aligned

# Potential Contributions

- Identify where the specific layers where geospatial representations are most concentrated, moving from semantic tokens to physical coordinates.
- **Geospatial Activation Steering (GAS):** introduce a technique to steer the LLM's output by applying vector offsets without changing prompting
- Show that steering the geospatial vectors significantly reduces spatial hallucinations and improves performance on regional downstream tasks

# Research Gap: How does the model represent the *concept* of space, and can we manually manipulate it?

- **The Causal Aspect**
- Show that geospatial representations are not just "metadata" but are **causally linked** to the model's reasoning.
  - previous work showed the model *has* a map but I want to show that shifting the internal "coordinate vector," can force the model to change its factual output
  - (e.g., changing its description of local flora, climate, or infrastructure) without changing the text prompt.
- **Steering Continuous Spaces**
  - **Geospatial Activation Steering (GAS)**, a framework for manipulating continuous, multi-dimensional coordinate gradients within the transformer.
  - Most "Representation Engineering" focuses on binary behaviors but this is a new approach to steering numerical, spatial gradients It will show how to steer a model 500 miles North or to a specific arid climate zone using activation patching.
- **How it is better than just Prompting**
  - Show that internal vector steering is significantly more robust than prompt engineering for geographic metadata matching
  - when a model is "steered" toward a location, it is less likely to "spatial hallucinate" or be distracted by conflicting information in the prompt

# Impact

- Show that by steering the Spatial Vector, we can make the model understand the topography more clearly, leading to better disaster reasoning than if we just typed "Assume there is a mountain here" in the prompt.

# Background Literature

- <https://aclanthology.org/2025.emnlp-main.813.pdf>
  - In this work, we show that question ambiguity is linearly encoded in the internal representations of LLMs and can be both detected and controlled at the neuron level.
- <https://arxiv.org/html/2410.14155v2>
  - leverage a causal mediation technique called activation patching, to measure the faithfulness of an explanation towards supporting the explained answer
- <https://arxiv.org/pdf/2601.12626>
  - We search for such confluence, and ask whether the identified representation can causally explain aspects of input-output model behavior through a linear model.